Targeted Deep Survival Contrasts: Valid Inference for Individualized Treatment Benefit with Neural Networks

David McCoy[1] Yi Li[1] Mark J. van der Laan[1]

[1]Division of Biostatistics, University of California, Berkeley

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Abstract

Clinical deployment of neural survival models increasingly requires *predictive* claims—how much a treatment would change survival—not just prognostic risk scores. Such claims demand valid inference for counterfactual survival contrasts under confounding and covariate-dependent censoring, targets for which standard deep survival estimators are biased and provide no honest uncertainty. Building on Targeted Deep Architectures (TDA), which embeds TMLE in a network's weight space, we target the full vector of treatment-specific survival curves $\{S_1(t), S_0(t)\}$ over a time grid, and hence the benefit curve $S_1(t) - S_0(t)$ and the restricted mean survival time (RMST) difference. A single universal targeting update—a ridge projection of the stacked efficient influence functions onto closed-form last-layer gradients—debiases every coordinate at once, and a multiplier bootstrap yields simultaneous confidence bands for the whole benefit curve. In 200-replication simulations with confounded treatment, sign-varying effect heterogeneity, and censoring that depends on treatment and covariates, our estimator attains nominal pointwise coverage (95.6%) and near-nominal simultaneous band coverage (94.0%), with benefit-curve bias roughly a third of—and intervals 8% narrower than—a per-timepoint one-step (AIPCW) correction built from the same nuisance fits at comparable coverage; in a companion misspecification study an isotonized one-step closes essentially none of the corresponding gap, so the gain reflects targeting itself rather than monotonicity. The same study delineates the method's scope: point-estimate double robustness largely survives a badly misspecified outcome network—bias an order of magnitude below the plug-in, though with inflated variance—but the working-model intervals then undercover while the standard one-step stays valid, quantifying the price of superefficiency and when to prefer each estimator.

1 Introduction

Neural networks are now standard tools for individual-level survival prediction from rich clinical inputs, including digital pathology and other high-dimensional modalities. Clinical decisions, however, hinge on *contrasts*: how survival would differ under one treatment versus another. Estimating such contrasts from observational data raises two familiar obstacles—confounding and outcome-dependent censoring mechanisms—and a less appreciated third: a network trained for predictive loss is a biased, inefficient estimator of any low-dimensional causal functional [3, 10], and its naive standard errors are invalid.

Semiparametric theory offers the remedy: debias a flexible initial fit by solving the efficient influence function (EIF) equation, via one-step/AIPCW corrections [3, 6] or targeted maximum likelihood estimation (TMLE) [11]. For survival targets these tools are mature [8, 1, 9], but standard implementations correct *outside* the model: the one-step adds the empirical mean of the EIF to a plug-in, and classical TMLE fluctuates the output distribution after training. Both become awkward when the target is a whole *vector*—two survival curves on a fine grid—where per-coordinate corrections need not respect monotonicity and can push estimates off the model space. Targeted Deep Architectures (TDA) [7] instead embeds the TMLE fluctuation *inside* the network: a small targeting subset of weights is updated along the projection of the EIF onto per-sample loss gradients, so the fitted network itself is the debiased estimator. TDA was demonstrated for the average treatment effect and the marginal survival curve without treatment; the clinically decisive object—the causal contrast of treatment-specific survival curves under confounding and dependent censoring—was not addressed.

Related work. Neural estimators of treatment-specific survival curves exist—notably SurvITE [5], which targets exactly our estimand with balanced discrete-time hazard networks, and counterfactual time-to-event models [2]—but they provide point estimates without valid confidence statements; that is the gap we fill. Conversely, survival TMLEs with simultaneous inference exist [1, 9] but target the output distribution through analytic fluctuation submodels that become cumbersome for many coordinates and arbitrary architectures; TDA-style weight-space targeting is what removes that obstruction. Causal survival forests [4] give honest inference for conditional effects at fixed horizons rather than the full population benefit curve.

Contributions. We propose *Targeted Deep Survival Contrasts* (TDSC): (1) the target vector $\Psi = (S_1(t_1), \ldots, S_1(t_K), S_0(t_1), \ldots, S_0(t_K))$ with its stacked EIFs combining treatment and censoring weights (Section 2); (2) closed-form per-sample last-layer gradients for DragonNet-style hazard networks, making the universal TDA update for all $2K$ coordinates one ridge regression per iteration (Section 3); (3) simultaneous sup-t bands for the benefit curve via multiplier bootstrap; (4) simulations with strong confounding, qualitative effect heterogeneity, and treatment- and covariate-dependent censoring, in which TDSC attains nominal coverage with lower bias and narrower intervals than the per-timepoint one-step built from identical nuisance estimates (Section 5).

2 Setup, estimands, and efficient influence functions

Data. We observe n i.i.d. copies of $O = (X, A, \tilde{T}, \Delta)$: covariates $X \in \mathbb{R}^d$, binary treatment $A \in \{0, 1\}$, discrete follow-up time $\tilde{T} \in \{1, \ldots, K\}$, event indicator Δ. Time is coarsened into K bins; within a bin, events are recorded before censoring, and subjects surviving bin K are administratively censored. Let $Y(k) = \mathbf{1}\{\tilde{T} \geq k\}$ (at risk) and $dN(k) = \mathbf{1}\{\tilde{T} = k, \Delta = 1\}$ (event), and write $P_n f = n^{-1} \sum_i f(O_i)$. The likelihood factorizes through the propensity $g(X) = P(A = 1 \mid X)$, the event hazard $h(k \mid a, x)$, and the censoring hazard $h_c(k \mid a, x)$, with conditional survival $S(t \mid a, x) = \prod_{k \leq t} \{1 - h(k \mid a, x)\}$ and censoring survival S_c defined analogously.

Estimands. Under consistency, no unmeasured confounding, coarsening-at-random censoring (censoring may depend on A and X but not further on the event time), and positivity for treatment and censoring, the treatment-specific curves $S_a(t) = \mathbb{E}_X[S(t \mid a, X)]$, $a \in \{0, 1\}$, $t \in \{1, \ldots, K\}$, are identified. Our target is the $2K$-vector $\Psi = (S_1(\cdot), S_0(\cdot))$; the *benefit curve* $\beta(t) = S_1(t) - S_0(t)$ and the RMST difference $\Delta_{\text{RMST}} = \sum_{t=1}^{K} \beta(t)$ (bin width one; the contrast in expected bins survived through horizon K) are linear in Ψ, so their influence functions follow by linearity.

Efficient influence functions. For each coordinate $S_a(t)$ the EIF at P is the discrete-time analogue of Moore and van der Laan [8]:

$$D_{a,t}(O) \;=\; -\sum_{k=1}^{t} \frac{\mathbf{1}\{A=a\}}{g_a(X)\,S_c(k-1 \mid a,X)}\,\frac{S(t\mid a,X)}{S(k\mid a,X)}\Big(dN(k) - Y(k)\,h(k\mid a,X)\Big) + S(t\mid a,X) - S_a(t), \tag{1}$$

with $g_1 = g$, $g_0 = 1-g$. The first term is a weighted sum of martingale residuals—the within-bin "surprise" of an event relative to the fitted hazard—inversely weighted by the probability of receiving arm a and remaining uncensored through bin $k-1$; the second is the plug-in centering term. Stacking (1) over a and t gives the $n \times 2K$ influence matrix D whose column means $P_n D_{a,t}$ are exactly the first-order bias terms that targeting drives to zero.

3 Targeted Deep Survival Contrasts

Architecture and initial fit. We use a DragonNet-style network [10] adapted to discrete-time survival: a shared trunk feeding a propensity head for g and two outcome heads producing hazard logits $\{h(k \mid a,x)\}_{k=1}^{K}$ for $a \in \{0,1\}$, trained on the observed-arm discrete-time negative log-likelihood (masked per-bin binary cross-entropy) plus a propensity term, with early stopping. A separate network fits the censoring hazard. Any backbone—including a frozen foundation-model encoder with a trained head—can replace the trunk. Full architecture and hyperparameters are in Appendix A.

Targeting submodel and closed-form gradients. Following TDA, the targeting parameters ϑ are the final linear layers of the two outcome heads, $p = 2K(F+1)$ parameters for head width F. Because those layers are linear in the head features $\phi_a(X)$ and the loss is a per-bin Bernoulli likelihood, the per-sample gradient is available in closed form:

$$\frac{\partial \ell_i}{\partial W_a[k,\cdot]} = \mathbf{1}\{A_i=a\}\,Y_i(k)\big(h(k\mid a,X_i) - dN_i(k)\big)\,\phi_a(X_i), \tag{2}$$

and analogously for biases, giving the $n \times p$ score matrix G in one vectorized pass with no per-sample automatic differentiation.

Universal targeting update. Each iteration solves one ridge system for all $2K$ targets, $\alpha = (G^\top G + \lambda I)^{-1} G^\top D$ (with λ scaled to the mean diagonal of $G^\top G$; note $p > n$ here, so the ridge penalty is statistically active), giving per-target directions α_j and projected bias terms $d_j = P_n[G\alpha_j]$. The universal direction is $\alpha^* = \sum_j (d_j/\|d\|_2)\,\alpha_j$ [7, Sec. 2.3], applied as $\vartheta \leftarrow \vartheta - \gamma\,\alpha^*$ with a backtracking line search on $\sum_j (P_n D_j)^2$. After each step the hazards and influence matrix are recomputed (g and S_c stay fixed); iteration stops when every $|P_n D_j| \le \widehat{\mathrm{sd}}(D_j)/(\sqrt{n}\log n)$ or no step improves the criterion (in our simulations the tolerance was never met in full; the line search exited first in every replication—Section 5 reports the diagnostics). Two diagnostics are recorded: the relative projection residual $\|D - G\alpha\|/\|D\|$ (an empirical check of gradient coverage) and the final $\max_j |P_n D_j|/\mathrm{tol}_j$; any coordinate left above tolerance can receive its residual $P_n D_j$ as a one-step top-up, restoring an exactly solved EIF equation. The estimator is the plug-in from the targeted network—by construction a pair of monotone survival curves.

Cross-fitted variant. To satisfy the sample-splitting premise of the theory below without Donsker conditions on the network class, TDSC admits a V-fold cross-fitted form (CV-TMLE structure): for each fold, the nuisance networks are trained on the complement, and the targeting step solves the EIF equations *on the held-out fold*—gradients, influence functions, and plug-in all evaluated on data independent of the initial fit. Fold plug-ins are averaged; the pooled per-observation influence matrix supplies variance and bands. Both forms are implemented; Section 5 compares them empirically.

Inference and simultaneous bands. Coordinate standard errors are $\widehat{\mathrm{sd}}(D_j)/\sqrt{n}$ with D evaluated at the final fit; contrasts inherit influence functions by linearity. For the benefit curve we report a sup-t band: with i.i.d. standard normal multipliers ξ_i, the 95% quantile c^* of $\max_t |n^{-1}\sum_i \xi_i D_{\beta,t}(O_i)|/\widehat{\mathrm{se}}_t$ over bootstrap draws gives $\hat{\beta}(t) \pm c^* \widehat{\mathrm{se}}_t$ (c^* floored at 1.96, a conservative choice).

4 Theory

TDSC is TDA applied to a $2K$-dimensional pathwise-differentiable target, and its guarantees specialize those of Li et al. [7, Thms. 3.1–3.2] and adaptive debiased machine learning (ADML) [12]. We state the specialization for the *cross-fitted* implementation (Section 3; V-fold, nuisances trained out-of-fold, targeting and evaluation on the held-out fold), which satisfies the sample-splitting premise of those results without Donsker conditions on the network class. Let $M_n \subseteq \mathcal{M}$ be the working submodel induced by partially unfreezing ϑ, $\Psi_n(P_0) = \Psi(\Pi_n P_0)$ the working parameter (projection of P_0 onto M_n), and D_0 the $2K$-vector of EIFs (1) at P_0.

Assumption 1 (Identification and positivity). *Consistency, no unmeasured confounding, coarsening-at-random censoring; and $\delta \le g_0(X) \le 1-\delta$, $S_{c,0}(K-1 \mid a, X) \ge \delta$ a.s. for some $\delta > 0$.*

Assumption 2 (Nuisance rates). *The cross-fitted hazard and weight estimators satisfy $\|\hat{h} - h_0\|_{P_0}\big(\|\hat{g} - g_0\|_{P_0} + \|\hat{S}_c - S_{c,0}\|_{P_0}\big) = o_p(n^{-1/2})$, with all estimators respecting the bounds of Assumption 1.*

Assumption 3 (Gradient coverage). *The $L^2(P_0)$-projection of each $D_{0,j}$ onto the span of the final-layer scores $\nabla_\vartheta \ell$ has residual $\epsilon_{n,j}$ with $\|\epsilon_{n,j}\|_{P_0} = o_p(1)$ and $(P_n - P_0)\epsilon_{n,j} = o_p(n^{-1/2})$ for each j.*

Assumption 4 (EIF equations solved). $\max_j |P_n D_j| = o_p(n^{-1/2})$ *at the final fit (enforced exactly by the residual top-up).*

Theorem 1 (Joint asymptotic linearity). *Under Assumptions 1–4, the cross-fitted TDSC estimator satisfies, jointly in $\mathbb{R}^{2K}$, $\widehat{\Psi} - \Psi_n(P_0) = P_n D_0 + o_p(n^{-1/2})$, so $\sqrt{n}\,(\widehat{\Psi} - \Psi_n(P_0)) \rightsquigarrow N(0, \Sigma_0)$ with $\Sigma_0 = P_0 D_0 D_0^\top$. If additionally the ADML oracle-bias conditions hold ($\Psi(\Pi_n P_0) - \Psi(P_0) = o_p(n^{-1/2})$), the same expansion holds for the true $\Psi(P_0)$, and the limiting variance is no larger than the nonparametric efficiency bound (superefficiency when P_0 is well-approximated by M_n).*

Corollary 1 (Contrasts). *The benefit curve $\hat{\beta}(\cdot)$ and $\widehat{\Delta}_{\mathrm{RMST}}$ are asymptotically linear with influence functions $D_{1,t} - D_{0,t}$ and $\sum_t (D_{1,t} - D_{0,t})$, by linearity of both maps.*

Proposition 1 (Simultaneous band). *Under the conditions of Theorem 1, the Gaussian-multiplier estimate of the* 95% *quantile of* $\max_t |\mathbb{G}_t|/\sigma_t$, *for $\mathbb{G}$ the limiting Gaussian of Corollary 1, is consistent, and the sup-t band covers the whole working benefit curve with asymptotic probability* 0.95.

Proposition 2 (Top-up restores double robustness)**.** *The top-up estimator* $\widehat{\Psi}^+ = \widehat{\Psi} + P_n \widehat{D}$ *equals the AIPCW one-step evaluated at the targeted nuisances. Consequently, under Assumption 1 and standard convergence of the estimated nuisances to* some *limits,* $\widehat{\Psi}^+$ *is consistent for* $\Psi(P_0)$ *if either the hazard limit equals* h_0 *or the weight limits equal* $(g_0, S_{c,0})$ *jointly—regardless of whether targeting converged.*

Proof sketches are in Appendix B; they specialize the ADML arguments using the exact second-order remainder of the survival EIF (a telescoping identity), with cross-fitting controlling the empirical-process term.

Remark 1 (No-split implementation and scope)*.* Theorem 1 covers the cross-fitted variant; the no-split implementation is not covered as stated (it violates the sample-splitting premise of TDA's first-order expansion), and its validity at moderate n is an empirical matter—Section 5 finds nominal behavior at $n \geq 1000$. Two further caveats. Double robustness of the *point estimate* (Proposition 2) does not extend to inference: consistent variance estimation still requires the relevant nuisances, and Section 5 quantifies the failure modes. Superefficiency likewise cuts both ways: narrower intervals when M_n approximates P_0 well, undercoverage when it does not—the empirical price appears under outcome misspecification and at small n.

5 Simulation study

Design. Covariates $X \sim N(0, I_{10})$; treatment $A \mid X \sim \text{Bern}(\text{expit}(0.9X_1 - 0.7X_2 + 0.6X_3))$, truncated to $[0.05, 0.95]$; event hazard $h(k \mid a, x) = \text{expit}(-3.1 + 2k/K + 0.7x_1 + 0.5x_2 + 0.4x_3 - 0.3x_4 + a\,\tau(x))$ with $\tau(x) = -1 + 0.8x_1 + 0.6x_3$, so the treatment effect changes sign across the population; censoring hazard $\text{expit}(-3.6 + 0.5x_1 + 0.4x_2 + 0.4a + 0.3x_5)$, dependent on treatment and on covariates shared with the outcome (coarsening at random holds; unadjusted estimators are biased). We use $K = 30$, $n = 1000$ ($\approx$61% events, 47% treated in the demo replication). Ground truth is computed by exact integration over 4×10^5 covariate draws; the true Δ_{RMST} is 5.45 bins.

Estimators. (i) **Naive plug-in**: the initial network's curves; (ii) **unadjusted Kaplan–Meier** by arm; (iii) **one-step (AIPCW)**: per-timepoint $\widehat{\Psi}_j^{\text{plug}} + P_n D_j$ at the initial fit, using the same nuisances as TDSC; (iv) **TDSC**, plus its residual top-up variant; (v) in the scenario grid below, an IPTW×IPCW-weighted KM and an *isotonized* one-step (each arm projected onto monotone curves), which separates how much of TDSC's advantage is explained by monotonicity alone. Causal survival forests and an output-space survival TMLE [9] are the remaining planned comparators.

Results. Table 3 reports time-averaged operating characteristics over 200 Monte Carlo replications (per-cell coverage MC SE $\approx$ 1.5 points). Four findings. First, adjustment is essential: unadjusted KM's absolute bias (0.121) is more than 50-fold TDSC's (0.0021), and its average Δ_{RMST} misses the truth of 5.45 entirely. Second, the naive plug-in carries the regularization-induced bias that debiasing exists to remove (0.021; RMST bias -0.38). Third, on the benefit curve TDSC and the one-step both achieve nominal pointwise coverage (95.6% vs 95.1%), but TDSC does so with roughly a third of the bias (0.0021 vs 0.0057) and 8% narrower intervals (0.140 vs 0.151); the MSE point estimates favor TDSC (0.00149 vs 0.00173), though the *paired* replication-level MSE difference (-0.00024, MC SE 0.00026) is not resolved at 200 replications. The narrower-at-nominal-coverage combination reflects the working-model efficiency of Section 3; the comparison does not isolate weight-space targeting per se from TMLE-versus-one-step, which the isotonized one-step and output-space TMLE comparators in the full study are designed to decompose. Fourth, the two estimators trade off differently on the RMST functional: TDSC is essentially unbiased (0.007 vs -0.161) but has higher variance, giving the one-step the lower RMST MSE (0.53 vs 0.79); both cover at ≈95–97%. The simultaneous band covered the whole curve in 94.0% of replications, with $c^* \approx 2.9$. Targeting averaged 9.1 iterations. Figure 1 shows one replication (seed 7; representative—per-replication MAE rankings match the table).

Nuisance misspecification and mechanism. A second study (150 replications per arm; propensity fit by a standalone network throughout so the outcome and weight models can be blinded independently) stresses both arms of double robustness by hiding key covariates from one set of nuisances at a time (Table 2). Three conclusions. *Monotonicity is not the mechanism*: in the well-specified arm the isotonized one-step is indistinguishable from the raw one-step (MSE 0.00154 vs 0.00155), while in this configuration TDSC's paired MSE advantage is clearly resolved (-0.00052, MC SE 0.00009)—the gain comes from targeting, not from projecting curves onto the monotone cone. *Point-estimate double robustness holds, with a caveat*: blinding the outcome network to the main confounder and effect modifier explodes the naive plug-in (bias 0.19; RMST bias -5.8) while the EIF-corrected estimators stay an order of magnitude less biased; blinding the weight models instead leaves both corrected estimators protected through the hazard arm while the weighting-only IPTW×IPCW KM fails (bias 0.076).

Inference is not doubly robust, and TDSC pays first: under outcome misspecification TDSC's working-model intervals undercover badly (64.6%; top-up 84.3%) while the one-step—whose wider intervals track the nonparametric variance—stays at 95.0%. A *safeguarded* variant—the top-up point estimate paired with the nonparametric (initial-fit) SE—repairs much of the shortfall (coverage 88.8%) at the one-step's width, but only the one-step itself remains fully nominal, consistent with the targeted fit drifting along a badly misspecified gradient span: even the raw TDSC estimate paired with the nonparametric SE covers only 72%, so the failure lives in point-estimate variability that no IF-based SE captures. Under weight misspecification all corrected estimators undercover mildly (≈89%) through the variance estimate. Diagnostics flag the failure in-sample: the mean number of coordinates left above tolerance rises from 37 (well-specified; nominal coverage regardless) to 54 (outcome-misspecified), and the strict $\mathrm{sd}/(\sqrt{n}\log n)$ tolerance was never met in full in any of the 450 replications—the line search exits first—so we recommend reporting unsolved-coordinate counts and applying the residual top-up as defaults.

Sample-size scaling. Repeating the main (well-specified, Table 3 configuration) study at $n \in \{500, 2000\}$ (Table 1) shows the expected $\sqrt{n}$ behavior from $n = 1000$ up: TDSC's $\sqrt{n}\times$bias is flat (0.067 at n=1000, 0.062 at n=2000), $n\times$MSE falls (1.49 to 1.02, dipping below the one-step's 1.43—the superefficiency signature), and pointwise and simultaneous coverage hold at 95–97%. At $n = 500$ the ranking *reverses*: TDSC's bias (0.0140) exceeds the one-step's (0.0056) and its coverage drops (91.4% pointwise, 85.3% simultaneous) while the one-step stays near nominal (94.8%). This is the misspecification lesson in another guise—an underpowered initial fit, whether from blinded covariates or a small sample, erodes the working-model gains first—so at small n the one-step is again the safer instrument.

6 Discussion

TDSC turns a two-headed neural survival network into a valid estimator of population treatment benefit at the cost of a few ridge regressions after training, and the misspecification grid gives the method an honest operating envelope: prefer TDSC when the initial network fits well (narrower

Table 1: Sample-size scaling, benefit curve (well-specified, Table 3 configuration; 150/200/150 replications).

n	Estimator	Bias	$n\times$MSE	Coverage	CI width	Simult. band
500	one-step	0.0056	2.10	94.8%	0.232	—
	TDSC	0.0140	2.77	91.4%	0.198	85.3%
1000	one-step	0.0057	1.73	95.1%	0.151	—
	TDSC	0.0021	1.49	95.6%	0.140	94.0%
2000	one-step	0.0025	1.43	93.7%	0.096	—
	TDSC	0.0014	1.02	96.2%	0.093	96.7%

Table 2: Nuisance-misspecification grid, benefit curve ($n = 1000$, $K = 30$, 150 replications per arm; coverage MC SE $\approx$ 1.8pp). *outcome mis.*: outcome hazards blinded to x_1, x_3; *weights mis.*: propensity and censoring models blinded to x_1, x_2. Propensity is fit by a standalone network (not Table 3's trunk head), so well-specified cells are not directly comparable to Table 3. The isotonized row changes only the point estimate; the safeguard row changes only the interval (its nonparametric SE equals the one-step's, hence the equal widths).

Scenario	Estimator	Bias	MSE	Coverage	CI width
well-specified	one-step	0.0051	0.00155	94.5%	0.144
	one-step, isotonized	0.0051	0.00154	—	—
	TDSC	**0.0036**	**0.00103**	95.7%	**0.135**
outcome mis.	naive plug-in	0.1922	0.04110	—	—
	one-step	0.0135	**0.00345**	**95.0%**	0.197
	TDSC	0.0119	0.01860	64.6%	0.156
	TDSC + top-up	0.0265	0.01325	84.3%	0.156
	+ nonpar. SE (safeguard)	—	—	88.8%	0.197
weights mis.	IPTW$\times$IPCW KM	0.0760	0.00720	—	—
	one-step	0.0114	0.00111	89.4%	0.111
	TDSC	0.0111	0.00114	89.2%	0.111

intervals at nominal coverage, gains not attributable to monotonicity; $n \gtrsim 1000$ in our design), and fall back when diagnostics (unsolved-coordinate counts, projection residuals) suggest otherwise or the sample is small—the safeguarded top-up repairs most of the coverage loss, but under gross outcome misspecification, and at $n = 500$, only the one-step is fully reliable. The remaining work: (i) evaluation of the cross-fitted variant (Sections 3–4) across the misspecification and scaling grids, where the variance-estimation failures make its theoretical protection most relevant; (ii) an output-space survival TMLE comparator to complete the mechanism decomposition, additional sample sizes, and sensitivity to the ridge penalty, targeting-layer choice, and weight-truncation levels (Appendix A); (iii) *calibration-for-benefit*—targeting arm-specific curves within strata of the network's own predicted benefit, the estimand used to validate individualized-benefit models against randomized trials—and application on frozen pretrained encoders (e.g. pathology foundation models), which the last-layer construction supports directly.

Table 3: Benefit curve $S_1(t)-S_0(t)$, time-averaged over 30 points ($n = 1000$, $K = 30$; 200 replications, coverage MC SE $\approx$ 1.5pp). Bias is mean absolute bias. The paired TDSC−one-step MSE difference is -0.00024 (MC SE 0.00026).

	Bias	MSE	Coverage (ptwise)	CI width
Naive neural plug-in	0.0213	0.0026	—	—
Unadjusted KM	0.1212	0.0163	—	—
One-step (AIPCW)	0.0057	0.0017	95.1%	0.151
TDSC (ours)	**0.0021**	**0.0015**	95.6%	**0.140**
Δ_{RMST}: one-step	bias -0.161	MSE **0.53**	96.5%	2.87
Δ_{RMST}: TDSC	bias **0.007**	MSE 0.79	95.5%	2.69
TDSC simultaneous band	coverage 94.0% (whole curve)		band width 0.207	

A Reproducibility

All code, raw Monte Carlo stores, and the exact scripts behind every table are at `https://github.com/blind-contours/tdsc`. Networks: trunk $d \to 64 \to 64$ (ELU); propensity head $64 \to 64 \to 1$; per-arm outcome heads $64 \to 64 \to K$; censoring network $(d+1) \to 64 \to 64 \to (K-1)$. Training: full-batch Adam, lr 10^{-3}, max 500 epochs, early stopping (patience 25) on a 20% validation split, propensity loss weight 1.0. Targeting: ridge scale $\lambda = 0.01 \times \operatorname{mean} \operatorname{diag}(G^\top G)$; $\gamma_0 = 1$ with up to 12 halvings; max 50 iterations; tolerance $\widehat{\mathrm{sd}}(D_j)/(\sqrt{n}\log n)$. Truncation: $\hat{g} \in [0.05, 0.95]$, $\hat{S}_c \geq 0.05$, conditional survival floored at 10^{-3} inside (1). Bands: 1000 Gaussian-multiplier draws. No cross-fitting is used (Section 3). Truth: 4×10^5-draw exact integration.

B Proof sketches

Theorem 1. Write, per coordinate, the standard decomposition $\widehat{\Psi}_j - \Psi_{n,j}(P_0) = (P_n - P_0)\widehat{D}_j + P_0\widehat{D}_j + P_n D_{0,j} - P_n\widehat{D}_j + P_n\widehat{D}_j$. Assumption 4 makes $P_n\widehat{D}_j = o_p(n^{-1/2})$. For the drift term, the survival EIF admits an *exact* second-order remainder: using $S(k) = \prod_{j \leq k}(1 - h(j))$ and the telescoping identity $\prod_{k \leq t}(1-\hat{h}(k)) - \prod_{k \leq t}(1-h_0(k)) = \sum_{k \leq t} S_0(k-1)\left(h_0(k)-\hat{h}(k)\right)\hat{S}(t)/\hat{S}(k)$, one verifies $P_0\widehat{D}_j + \Psi_j(\hat{P}) - \Psi_j(P_0) = R_{2,j}$, where $R_{2,j}$ is a sum over $k \leq t$ of integrals of $\{\hat{h}(k) - h_0(k)\}$ against $\{\hat{g}\hat{S}_c(k-1) - g_0 S_{c,0}(k-1)\}/\{\hat{g}\hat{S}_c(k-1)\}$-type differences; under the positivity bounds of Assumption 1, Cauchy–Schwarz gives $|R_{2,j}| \lesssim \|\hat{h} - h_0\|_{P_0}(\|\hat{g} - g_0\|_{P_0} + \|\hat{S}_c - S_{c,0}\|_{P_0}) = o_p(n^{-1/2})$ by Assumption 2 (this display also proves Proposition 2: R_2 vanishes if either factor's limit is correct). For the empirical-process term, cross-fitting makes $\widehat{D}_j$ independent of the fold on which $(P_n - P_0)$ is evaluated, so conditional on the training folds, $(P_n - P_0)(\widehat{D}_j - D_{0,j}) = o_p(n^{-1/2})$ whenever $\|\widehat{D}_j - D_{0,j}\|_{P_0} = o_p(1)$, which follows from the nuisance consistency in Assumption 2 and the gradient-coverage residual control in Assumption 3 (the targeted fluctuation moves ϑ within the score span, so the perturbation of $\widehat{D}_j$ is controlled by the same projection). Summing folds yields the coordinate-wise expansion; joint convergence follows from the multivariate CLT applied to the stacked D_0. The extension to $\Psi(P_0)$ and the (super)efficiency statement are Theorem 6/7 of van der Laan et al. [12] specialized as in Li et al. [7].

Corollary 1 and Proposition 1. Both maps are bounded linear in Ψ, so asymptotic linearity transfers with the transformed influence functions. Given joint normality with consistently estimated

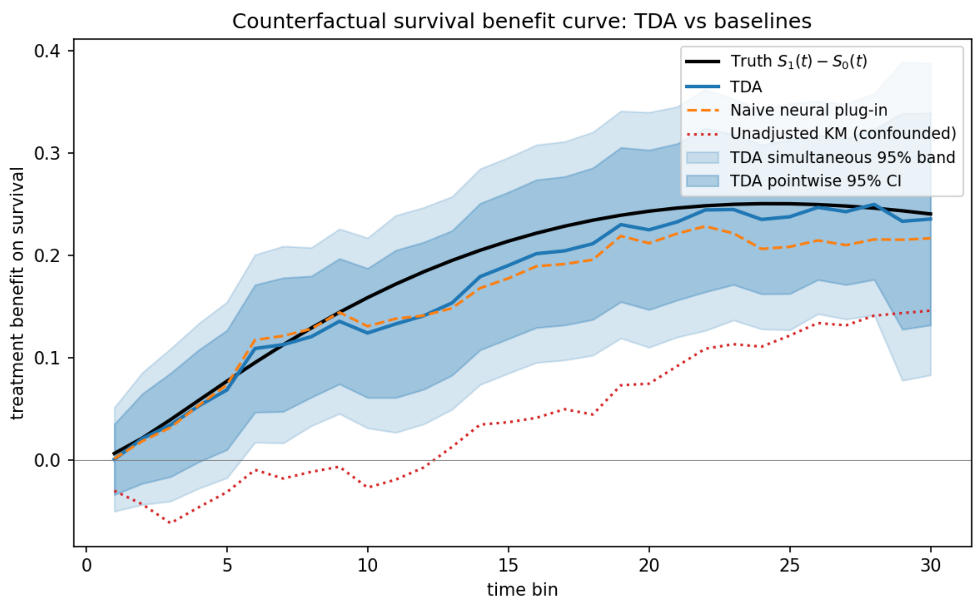

Figure 1: One replication ($n = 1000$, seed 7). The TDSC benefit curve tracks the truth, which lies inside both the pointwise CI and the simultaneous band; the naive plug-in drifts low at later times, and unadjusted KM is severely biased (wrong-signed at early times).

Σ_0 (IF second moments), the conditional multiplier process $n^{-1/2} \sum_i \xi_i \widehat{D}_{\beta,t}(O_i)$ converges weakly to the same Gaussian limit (conditional multiplier CLT), so its sup-quantile is consistent for the sup-quantile of the limit; the $c^* \geq 1.96$ floor only enlarges the band.

Proposition 2. By definition $\widehat{\Psi}_j^+ = P_n[\hat{S}(t \mid a, X)] + P_n \widehat{D}_j$, which is precisely the AIPCW one-step at nuisances $(\hat{h}^{\text{targ}}, \hat{g}, \hat{S}_c)$. Its population bias is the R_2 term above evaluated at those limits, which vanishes under either-arm consistency. □

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